Modeling Variations in State Education Attainment

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The study of educational outcomes using quantitative processes has a rich history, beginning with the 1966 Coleman Report, a foundational study that merged sociological investigation with statistical analysis. The report, which was born out of the necessity to establish a baseline understanding of differences in the educational experiences of black and white students, sought to identify which factors were most salient to predicting a student's outcomes through the recently desegregated education system in the United States. Coleman's findings spurred their fair share of controversy, as the report argued that (in Coleman's own words), "All factors considered, the most important variable—in or out of school—in a child's performance remains his family's education background" (Dickinson, 2016, para. 21). This claim rejected the commonly held belief that the achievement gap between students of color and white students was due to discrimination in resource distribution and disparate levels of teacher quality.

While the Coleman Report focused on a descriptive assessment of the state of education in the early years of the post-Brown v. Board of Education environment, its findings, as well as the numerous avenues of research inspired by the report, have been used to drive policy changes aiming to improve the quality of education in the United States. In the years that followed, researchers sought to define a mathematical model of education by adapting supply-side and demand-side models from economics; this push led to the genesis of education production functions that relate student's academic success to a variety of factors. The necessity of continued work in this field is clear, as there have been steep declines in student achievement measures in recent years. One need not look far to see that impact of the Covid-19 pandemic has undeniably affected the state of education worldwide. The latest results of the National Assessment of Educational Progress (NAEP) showed significant declines in reading and

mathematics in 2022 (Binkley, 2023a); similar declines were seen in the Programme for International Student Assessment (PISA) administered in 2022 (Binkley, 2023b). Importantly, these trends began before the disruptions caused by the pandemic, which further highlights the need for research in this vein.

Research Question

Two common lenses used to assess student outcomes are academic achievement, typically measured via standardized testing, and educational attainment as measured by level of education completed. There exists a plethora of data regarding education in the United States, a fair amount of which is accessible for public use via the National Center for Education Statistics (NCES) and the NAEP's data exploration tool, known as The Nation's Report Card; likewise, a variety of socioeconomic factors are reported regularly through the US Census Bureau and Congressional Research Services (CRS). During the initial exploration of the 2022 NAEP results, it became clear that there were significant variations in student outcomes based on where students lived. This geographical variation also bore out in attainment rates, especially at the college/no college threshold. These findings led to the formulation of the following research question: "What state-level characteristics drive educational attainment differences within the United States?"

Information

To answer this question, a dataset was compiled that sought to capture a few key components which appeared in the literature. Maurice Boissiere's 2004 report for the World Bank, "Determinants of Primary Education Outcomes in Developing Countries" proved particularly useful in guiding the framework for data collection. Despite the focus on developing

countries, the report's summary of research often builds on studies completed within developed countries, providing a sort of bird's eye view on contemporary debates over which factors were most salient as predictors. Boissiere's report identified the following categories for consideration:

> (1) "hardware" such as school buildings, classrooms and furniture, sanitation, etc., (2) "software" such as curriculum, pedagogy, textbooks, writing materials, etc., and (3) teachers. A fourth category of management and institutional structure and a fifth category of context and background variables (student academic ability, family, and community background, etc.) could be added as well. The management category looks at how allocation decisions are made regarding the first three categories of inputs. (p. 15)

With this set of potential categories in mind, data was pulled from several elements of NCES to address these areas. In the broadest sense, the predictor variables could be grouped into state's per capita expenditures on education, teacher pay and quality (measured by first year certification requirements), pupil to teacher ratio, early education emphasis, state poverty rates and state's political lean. There are not currently databases to track the "software" category in an efficient manner, and this gap is one area that begs for additional investigation. The difficulties in gathering such data are inherent to the nature of the state based educational system, as decisions like which curriculum to adopt typically occur at the county or district level. Attempts to make federal or even state level mandates with regards to curriculum can ignite opposition, as can be seen with the rollout of Common Core Standards that are now widely adopted (Bidwell, 2014).

The inclusion of a political lean factor in this dataset was a decision not taken lightly. This researcher once likened it to "licking a lightning rod during a thunderstorm," (the one time I will quote myself, to add a drop of humor but also to illustrate my strong desire NOT to include any political factors, due to the inherent messiness it could cause while interpreting results).

Third rail imagery aside, the purpose of including the lean factor was twofold – it serves as a crude shorthand for the manner that state and federal dollars are allocated (addressing Boissiere's fourth category), and it provides another indicator of the broader social context of the state (the fifth category). David Griffith, a senior director with the Association for Supervision and Curriculum Development, summarized the two parties' policy platforms as follows: the Republican platform around education is focused on increased school choice, a decrease in overall expenditures and federal interventions, a rejection of Common Core Standards and return to conservative values based curricula, and a push towards skills and vocational training instead of the traditional four-year university education as the typical pathway for secondary education. On the other hand, the Democratic platform is focused on improving public schools, increased teacher pay and recruitment, a focus on STEAM education in K-12 classes, and improved access to higher education as the typical pathway for secondary education (Griffth, 2020).

Assessment

Given the constraints of the course, several limiting decisions were made to ensure that a suitable model could be produced in a reasonable timeframe. The first limiter was to create a model that captured a snapshot in time, with a focus on the latest information available. This meant that potential comparative studies with a pre/post-Covid (or Common Core for that matter) approach were no longer achievable, but this simplified the construction of the dataset immensely, as some factors, like teacher certification and kindergarten requirements, were only available for the latest reporting year. The second limitation imposed by the frame of the research question is the non-inclusion of student learning capacity as a factor, and the wholesale sidelining of factors like student race and sex in the aggregate. These are undoubtedly important variables for individual student outcomes; disregarding them would be non-trivial in a

longitudinal study or an exploration of the effects of classroom composition on student achievement. However, this study is focused on the interactions between state level policies and (primarily) economic factors. The model discussed in the next section is not intended to make predictions about a particular student's outcomes, but rather, to make predictions about how educational attainment might change if a state adjusted policies around education and spending.

Several individual factors have baked-in assumptions that should be addressed. The requirements around kindergarten offering and attendance are current as of 2017-18, but the cadre of NAEP test takers for 2022 would have been past kindergarten age at the time of these requirements reporting. Since an earlier version of this report was not available, there is a chance that some state's data is not reflective of the environment the cadre in question would have experienced. Similar issues arose from the number of minimum hours of instruction that students received by 8th grade, although the impact is less intense as at most only 4 years of instructional time could be incorrect, and if anything, those values would be slightly below their actual amount. Several states do not set minimum hours required, which is a source of potential issues as well, one that almost led to the factor not being included in the final model. There is also an implicit assumption made in the minimum hours calculation that this value is for a student that stays within the same state for the entirety of their K-8 education. In terms of educational attainment rates, there is no accounting for how much of that state's percentage of high school diploma or bachelor's degree holders attained said diploma/degree in-state versus out of state.

All data gathered was ported into a single Excel spreadsheet named MAT430_Education. Three sheets were generated, one with all non-political variables included for all fifty states and District of Columbia, another includes all data but eliminates any observations with unreported variables, and the sheet that was ultimately used in to build the model, which includes all

variables for all fifty states but no DC observation. The District of Columbia's unique status as the country's capital meant that it was an outlier in nearly every category; it also seemed imbalanced to not consider other territories like Puerto Rico and Guam while including DC. Within the "all_data_sans_dc" data sheet, there are 50 observations and 28 factors (not including location name). A full summary of factors is included in the appendix.

Model

The first step in the model's creation was a correlation analysis performed using the programming language R and the integrated development environment RStudio. Two types of models were considered at this point in the analysis. The first was a pair of logistic regressions that classified states based on whether they met/exceeded the national average on the 2022 NAEP Reading and Mathematics exams. The second was a multiple linear regression model that used educational attainment or average NAEP scores as its response variable. With both cases, a series of data frames were formed that bundled related factors together, then the R command *pairs()* was used to generate correlation plots against the selected response variables. In addition to teasing out insights into which factors seemed to have the strongest correlations, this visual analysis provided insight into potential impacts of outliers and missing values. There were several predictor factors that showed strong collinearity, and a few potential quadratic relationships appeared as well. Generally speaking, the percentage of a state's population that earned a bachelor's degree was the most responsive of all potential response variables.

Selection

With the strongest predictors identified for both model types, simple one variable logistic and linear regressions were performed. Due to the differences in the purpose of the two model

types, a direct comparison of their performance is not possible. That said, the simple logistic model's classification power proved very limited, with the best of the logistic prototypes achieving poor results on their Receiver Operating Characteristic Curves. The typical Area Under the Curve for all logistics models were well under 40%, indicating that the models' performance classifying whether states met/exceeded national averages was lackluster at best.

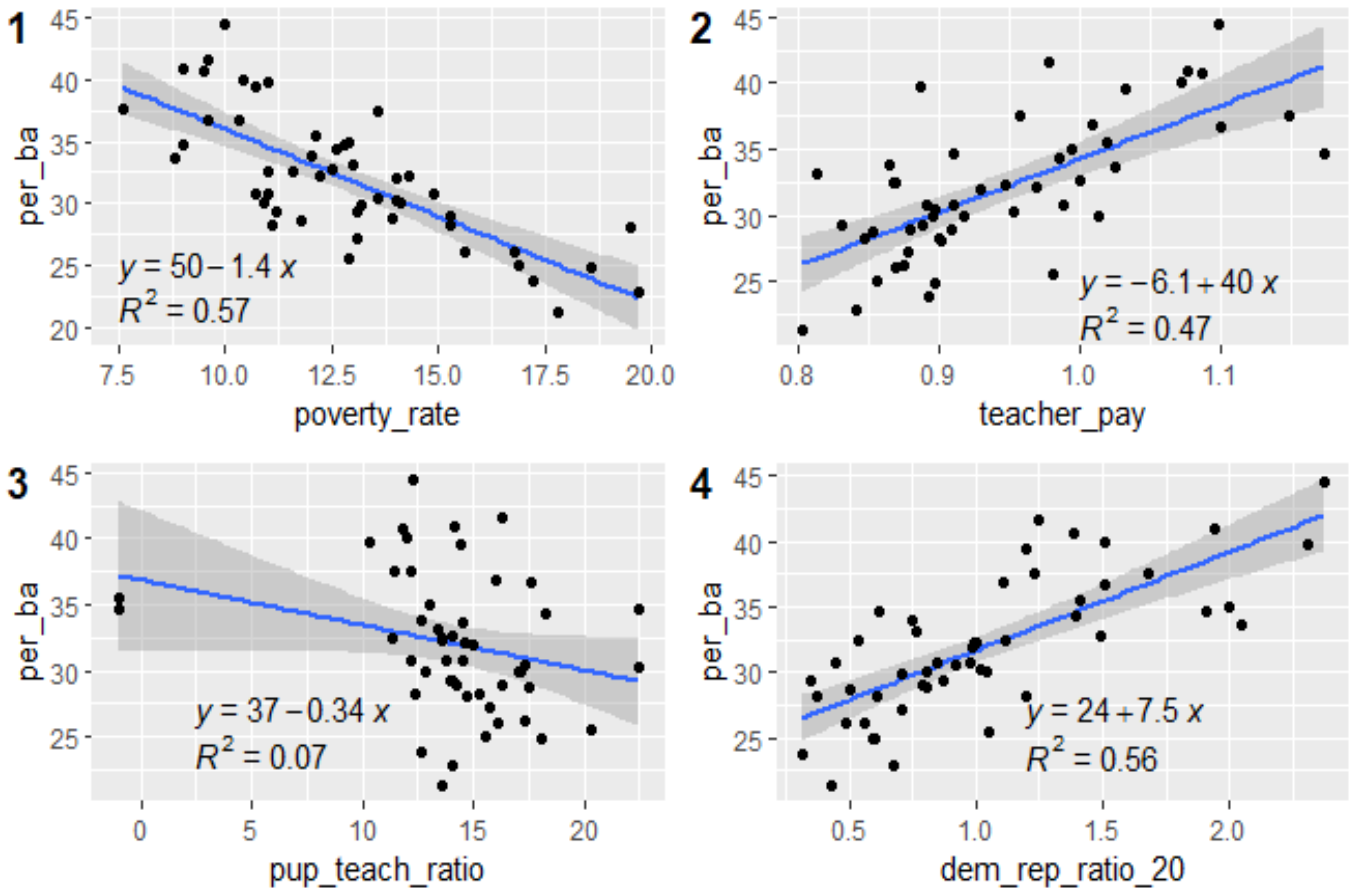
The results of the simple linear regressions came back somewhat stronger, with a few of the single variable prototypes eclipsing an R^2 value over 0.5, indicating that the simple models were capable of accounting for at least 50% of the variation observed within the data. Between that fact and the signs of strong linear correlations between the percent of a state's population with a bachelor's degree, it followed that a multiple linear regression would be the most appropriate model choice to investigate the research question discussed earlier. In addition to the "better" performance relative to the logistic regression, a multiple linear regression could be used to both identify the most impactful factors and make predictions about the potential impacts policy changes could have in the future. This model choice is well-aligned to typical education production functions, which is a step beyond the constraints of this paper's model, but certainly the closest analog.

Of the potential predictors, there were four factors that consistently registered as statistically significant at the 0.05 level (at least, though most were under 0.01).

poverty_rate	State's percent of population living below federal poverty level (2018 estimates from CRS)
teacher_pay* *misses the 0.05 cutoff by 0.0028	State's average teacher salary normalized with the national average set as 1 (CWIFT data from 2020)

pup_teacher_ratio	State's average pupil to teacher ratio (NCES data from 2020)
dem_rep_ratio_20	State's ratio of total votes cast for democrats to total votes cast for republicans (total as in state totals for all three offices, president, house, and senate) in 2020.

The first set of correlation plots (included below) show the dynamics of the variables that made it into the final model; each predictor is plotted against the percent of the state that has attained a bachelor's degree. A second set of correlation plots in the appendix shows the predictor variables plotted against each other; these were used to assess multicollinearity, further discussed later.



Creation

A linear regression model with four terms has the general form:

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

The prediction equation for this study's model is:

$$\hat{y} = \widehat{\beta_0} + \widehat{\beta_1} x_1 + \widehat{\beta_2} x_2 + \widehat{\beta_3} x_3 + \widehat{\beta_4} x_4$$

Where $\hat{y}$ is the predicted percent of population with a bachelor's degree; x_{1-4} are state level factors. Specifically, x_1 is the poverty rate, x_2 is teacher pay, x_3 is pupil to teacher ratio, and x_4 is political lean. $\widehat{\beta_0}$, $\widehat{\beta_1}, \widehat{\beta_2}$, $\widehat{\beta_3}$, $\widehat{\beta_4}$ are estimates of $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ respectively. R provides the following summary for this model:

```
Call:
lm(formula = per_ba ~ poverty_rate + teacher_pay + pup_teach_ratio +
    dem_rep_ratio_20, data = final_frame)

Residuals:
    Min      1Q  Median      3Q     Max
-6.7583 -1.2509 -0.1299  1.2407  5.9611

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)      30.32452    5.52027   5.493 1.75e-06 ***
poverty_rate     -0.85539    0.14150  -6.045 2.67e-07 ***
teacher_pay      11.68361    5.87602   1.988 0.052875 .
pup_teach_ratio  -0.17720    0.08783  -2.018 0.049626 *
dem_rep_ratio_20  3.98547    0.98165   4.060 0.000193 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.355 on 45 degrees of freedom
Multiple R-squared:  0.8151,   Adjusted R-squared:  0.7986
F-statistic: 49.58 on 4 and 45 DF,  p-value: 6.232e-16
```

Using the estimates from this output, the model's prediction equation comes out to:

$$\hat{y} = 30.3245 - 0.8554x_1 + 11.6836x_2 - 0.1772x_3 + 3.9855x_4$$

Model assessment and diagnostics are discussed in the Analysis section, but first, the process of reaching this model is addressed.

Process

With any multiple linear regression model, there is substantial risk of overfitting the data, especially in a scenario where the number of observations is relatively low. To prevent any potential overfitting, a series of nested F-tests were completed using R's *anova()* function. As each of the four factors was folded into the model, the new model's summaries were compared using three versions of the R^2 measure – traditional, adjusted, and predictive. Special attention was paid to the adjusted and predictive measures, as they are less susceptible to the artificial inflation that traditional R^2 suffers from by inclusion of additional variables. Those results, along with each model's Residual Standard Error (RSE) are summarized below.

Model	Multiple R-squared	Adjusted R-squared	Predictive R-squared	Residual Standard Error
lm1	0.5707	0.5617	0.5339	3.475
lm12	0.7248	0.7131	0.6878	2.811
lm123	0.7473	0.7308	0.6943	2.723
lm1234	0.8151	0.7986	0.7694	2.355

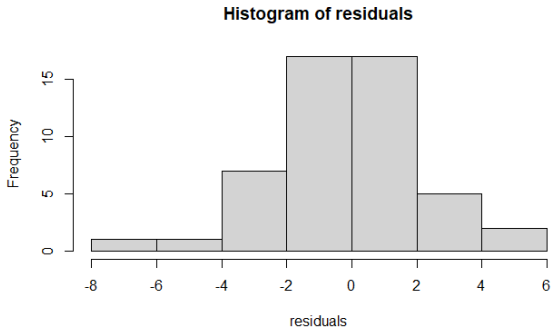
This table shows that model performance improved as each variable was added to the regression, with the three R^2 measures maintaining a tight difference throughout. The ANOVA test shown below confirms these results, indicating the inclusion of each additional variable should occur.

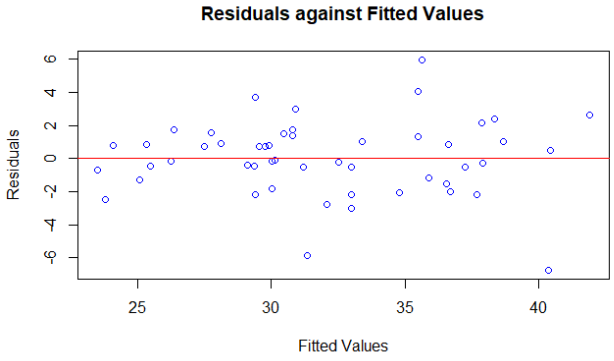
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Analysis of Variance Table

Model 1: per_ba ~ poverty_rate
Model 2: per_ba ~ poverty_rate + teacher_pay
Model 3: per_ba ~ poverty_rate + teacher_pay + pup_teach_ratio
Model 4: per_ba ~ poverty_rate + teacher_pay + pup_teach_ratio +
dem_rep_ratio_20
  Res.Df    RSS Df Sum of Sq       F    Pr(>F)
1     48 579.50
2     47 371.43  1   208.075 37.5104 2.039e-07 ***
3     46 341.06  1    30.371  5.4751 0.0237846 *
4     45 249.62  1    91.435 16.4833 0.0001935 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

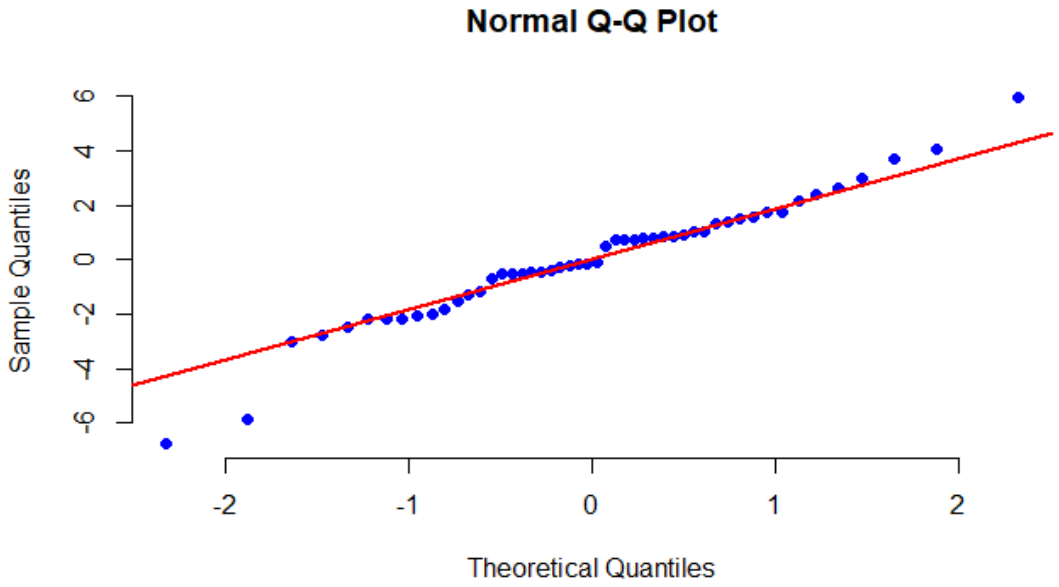
Analysis

To begin the model analysis, we start by assessing some standard diagnostic plots. A histogram of the model's residuals is used to determine whether the residual error is distributed normally, and the related Residuals against Fitted Values (RaFV) graph is used to verify assumptions about linearity and homoscedasticity. The final diagnostic plot, a Normal Q-Q plot, verifies normality if the plotted values tightly fit against the normal line.





The histogram shows a typical normal distribution with a slight leftward skew. There is a fairly balanced distribution about the zero-line in the RaFV graph with some imbalance at the upper



end of the fitted values. Since there is not significant evidence of funneling or fanning in the RaFV graph, it appears that the assumption of homoscedasticity holds. While there are some large departures at the upper and lower limits of the Normal Q-Q plot, the close fit of points within ± 1.5 theoretical quantiles suggests that the assumption of normality should hold, especially when considered in conjunction with the histogram of residuals.

One potential issue that arose with this set of factors is collinearity between several of the predictors. As the correlation plots included in the appendix show, there is evidence of linear relationships between teacher pay and political lean, teacher pay and poverty rates, and poverty rate and political lean. To better quantify this, Variance Inflation Factor (VIF) for all factors was calculated using R's *vif()* function, which is summarized below.

Factor	poverty_rate	teacher_pay	pup_teach_ratio	dem_rep_ratio_20
VIF	1.416493	2.404477	1.132228	2.304779

As none of the factors exceeded 2.5 (the lowest threshold discussed in Choueiry's article "What is an Acceptable Value for VIF?"), the strength of correlation between factors is low enough to justify retaining all four. The most conservative threshold was selected to counterbalance the low number of observations in the dataset.

These diagnostics reveal a model that handles the given dataset well. The core assumptions of multiple linear regression appear to be satisfied. Thus, we can turn to interpretation of the results and attempt to answer the research question formulated earlier. Recall that the prediction formula generated by R's beta estimates is:

$$\hat{y} = 30.3245 - 0.8554x_1 + 11.6836x_2 - 0.1772x_3 + 3.9855x_4$$

Where $\hat{y}$ is the predicted percent of population with a bachelor's degree; x_1 is the poverty rate, x_2 is teacher pay, x_3 is pupil to teacher ratio, and x_4 is political lean.

Based on this model, the simplest answer is that the four variables listed above are the biggest drivers of variation between state's education attainment rates. There are some serious caveats to this statement though, the first being these were the most impactful out of the seventeen potential predictors gathered during the early weeks of investigation. The list of potential unobserved confounders is immense, and more robust sensitivity analysis using R's *sensemakr* library would help define this issue. Further, the interaction between a state's political lean and teacher pay is a potential source of error that decreases the model's overall efficiency. While the teacher quality and early education emphasis factors did not make the cut individually, a primary component analysis might reveal that there was a stronger relationship than the discrete factors indicated. Another extension that could test the veracity of this model is applying it to datasets that capture alternate time frames. In fact, if there were opportunities to expand this work, these two steps would be the first extension of future model development.

A slightly more nuanced take on this equation requires additional discussion about the nature of these factors, specifically teacher pay and political lean. These two factors have the greatest magnitude of beta estimates, which might lead one to interpret them as the most important of the four factors in the model. However, the teacher pay variable does not represent a gross dollar amount or simple average salary, it is a labor index that compares several geographic and socioeconomic factors between each state. To say that the obvious solution is to pay teachers more would be an inaccurate interpretation; this is backed by a large volume of work done by the economist Eric Hanushek, who states in his 2020 essay "Education production functions*"* that there is "very weak support for the notion that simply providing higher teacher salaries or greater overall spending will lead to improved student performance" (p.165). Scatterplots of the per capita expenditures and teacher pay included in the MAT430_Education dataset seem to align

with Hanushek's argument, as there appear to be diminishing returns past a certain level of spending. Boissiere posits that a potential explanation for why expenditures appear to have strong effects in some regions while not in others has to do with outsized impact additional spending has in poorer areas, while richer environs are less responsive to extra dollars (2004). Thus, while increasing teacher pay might dramatically impact areas with weaker economic indicators, the same might not hold for wealthier zip codes.

On the question of political lean, the primary caveat is that this factor is the almost entirely out of the state's control; harkening back to the earlier discussion about this factor, it is meant only as a shorthand for addressing the categories Boissiere identified. A policymaker might use this model to predict how a policy aimed at reducing poverty, the average pupil to teacher ratio, or adjusting teacher pay would impact their state's education attainment rates. There is no use-case where said policymaker could adjust the political lean of the state, that would be a nonsensical application. Furthermore, a state shifting to the right end of the political spectrum does not translate directly to a decrease in bachelor's attainment rates, nor does a shift to the left translate to increased bachelor's attainment rates.

Ultimately, this model represents another entry in the growing body of research born out of the investigations inspired by the 1966 Coleman report. The model makes the case that the interplay between social, political and economic factors drive the differences in state education attainment, but it is far from a definitive accounting of all that goes into the complex machine that is the US primary education system.

Appendix

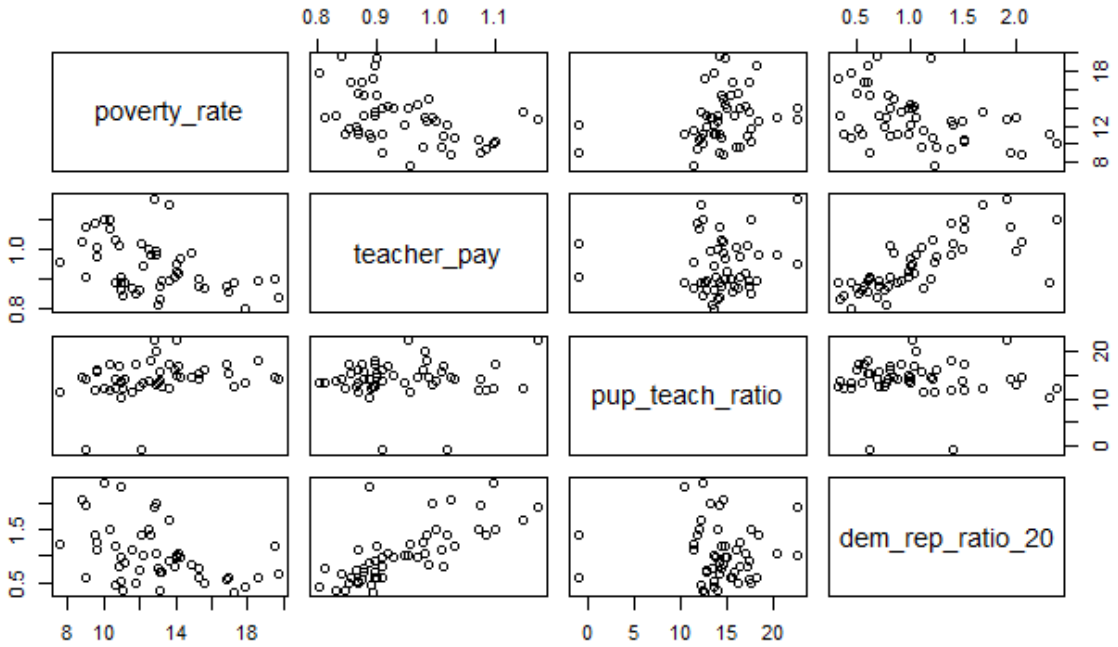
Table 1 – List of Variables in the MAT430_Education Dataset

VAR NAME	DESCRIPTION	Source
loc	Location name	
per_hs	Percent of high school graduates by 25	US Census via World Population Review
per_ba	Percent of bachelor's holders by 25	
read_avg	Average score on NAEP 2022 Grade 8 Reading exam	NAEP via Nation's Report Card
read_moe_navg	Did state meet or exceed national average on NAEP READ?	
Read_per_basic	Percent of students meeting a "basic" level or above on NAEP READ	
read_per_prof	Percent of students meeting a "proficient" level or above on NAEP READ	
math_avg	Average score on NAEP 2022 Grade 8 Math exam	
math_moe_navg	Did state meet or exceed national average on NAEP MATH?	
Math_per_basic	Percent of students meeting a "basic" level or above on NAEP MATH	
math_per_prof	Percent of students meeting a "proficient" level or above on NAEP READ	
both_moe_navg	Did state meet or exceed national average on BOTH NAEP exams	
percap_total	State's total spending per capita	Tax Policy Center
percap_prim_ed	State's spending on primary education per capita	
percap_higher_ed	State's spending on higher education per capita (post K-12)	
teacher_pay	Teacher's pay on normalized scale where 1 represents national average	NCES EDGE
pup_teach_ratio	Average pupil teacher ratio	NCES ELSI
fte_teacher_total	Total number of full-time equivalent teachers in 2021	
min_hours_by8th	Minimum number of instructional hours by 8th grade (assuming no churn)	NCES SEP

mand_attend_age	State's compulsory attendance age	
dist_req_k_offered	Whether a state requires districts to offer kindergarten	
k_attend_req	Whether a state requires kindergarten attendance	
knowledge_exam	Whether a state requires a knowledge exam for first year certification	NCES SEP
teaching_skill_exam	Whether a state requires a teaching skill exam for first year certification	
dem_rep_ratio_16	State's ratio of total votes for democrats to total votes for republicans in 2016	FEC
dem_rep_ratio_20	State's ratio of total votes for democrats to total votes for republicans in 2020	FEC
dtr_shift	Shift in ratio of votes from 2016 to 2020	derived from above
percap_title1_funds	State's per capita Title 1 funding	US Census via Journalist Resource (.org)
poverty_rate	Percent of state population living under the federal poverty line	US Census

Note: Sources grouped by color to indicate instances where multiple parameters were pulled from the same source.

Table 2 – Predictor Variables Correlation Plots



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